

Course Specification

(Bachelor)

Course Title: Research Method and Pharmacoepidemiology
Course Code: PHP 417
Program: PharmD
Department: NA
College: Pharmacy
Institution: Shaqra
Version: 2
Last Revision Date: 26/10/2023

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A. General information about the course:

1. Course Identification

1. Credit hours: (3)					
2. Course type					
A.	☐University	☒ College	☐Department	☐Track	☐Others
B.	☒ Required		☐Elective		
3. Level/year at which this course is offered: (7/4)					
4. Course General Description:					
This course aims to explore and integrate concepts and considerations to the conduct of pharmacoepidemiologic research. Topics covered include using automated databases, pharmacoepidemiological studies, the scope of biostatistics and Research methodology and important hypothesis-testing tools in evaluating post-marketing drug safety. Select and evaluate appropriate tools for the assessment of pharmacoepidemiological studies. Also gain the knowledge and skills they need to undertake a research project, to read and understand the scientific paper and their statistical significance.					
5. Pre-requirements for this course (if any):					
Biostatistics and data analysis (PHS 225)					
6. Co-requisites for this course (if any):					
7. Course Main Objective(s):					

1. To develop an understanding of pharmacoepidemiological studies, the scope of biostatistics and Research methodology and important hypothesis-testing tools in evaluating post-marketing drug safety.
2. A brief understanding of selecting and evaluating appropriate tools for assessing pharmacoepidemiological studies.
3. To empower the students with the knowledge and skills they need to undertake a research project, to read and understand the scientific paper and its statistical significance.

2. Teaching mode (mark all that apply)

No	Mode of Instruction	Contact Hours	Percentage
1	Traditional classroom	45	100%
2	E-learning		
3	Hybrid • Traditional classroom • E-learning		
4	Distance learning		

3. Contact Hours (based on the academic semester)

No	Activity	Contact Hours
1.	**Lectures**	45
2.	**Laboratory/Studio**	
3.	**Field**	
4.	**Tutorial**	
5.	**Others (specify)**	
Total		

B. Course Learning Outcomes (CLOs), Teaching Strategies and Assessment Methods

Code	Course Learning Outcomes	Code of CLOs aligned with the program	Teaching Strategies	Assessment Methods
1.0	**Knowledge and understanding**			
1.1	**Describe** the different terms of study design in Pharmacoepidemiology and their definition and applicability.	**K1**	• **Lectures and discussion**	• **Mid and final Exams**
1.2	**Demonstrate** the study design for the research problem considered for the study and will be able to frame the research questions.	**K2**		
2.0	**Skills**			
2.1	Apply the research methods.	**K1**	• **Lectures,** • **Class discussions,** • **Solving problems.**	• **Midterm and final exams.** • **Homework** • **Quizzes.**
2.2	Solve the problem and interpret the outcome of the study statistically.	**K2**		
2.3	Perform a collection of data and justify disease spread or treatment efficacy**.**	**K3**		
3.0	**Values, autonomy, and responsibility**			
3.1	Estimate the responsibility and ability to work as a team in various settings of scientific research labs. Analysis reporting.	**V1**	• **Group assignments** • **Small group work and discussion**	• **Team project** • **Oral exams.**

C. Course Content

No	List of Topics	Contact Hours
1.	Introduction to Pharmacoepidemiology.	3
2.	Epidemiological Concepts and Measures (1).	3
3.	Epidemiological Concepts and Measures (2).	3
4.	Formulating a research question in Pharmacoepidemiology: Variables and Statistics	3
5.	Observational Research Methods: Case Reports/Series and Cross-sectional Studies **(1).**	3
6.	Observational Research Methods: Case Reports/Series and Cross-sectional Studies **(2)**	3
7.	Observational Research Methods: Case-Control Study	3
8.	Observational Research Methods: Cohort Study	3
9.	Quasi-Experimental Design (Time-Series Analysis); Experimental Design (Efficacy Trials)	3
10.	Experimental Study Design (Pragmatic Trials)	3
11.	Systematic Reviews and Meta-Analyses	3
12.	Pharmacoeconomics: Cost-Minimization and Cost-Effectiveness Analyses	3
13.	Health-related Quality of Life (HR-QOL) and Humanistic Outcomes	3
14.	Application of Pharmacoepidemiology in Drug Surveillance and Safety Monitoring	3
15.	Drug Safety Monitoring	3
Total		**45**

D. Students Assessment Activities

No	Assessment Activities *	Assessment timing (in week no)	Percentage of Total Assessment Score
1.	Class activities /Assignments, homework, research, presentation, participation (in-class quizzes)	weekly	20%
2.	Midterm exam 1	$7^{th} – 8^{th}$	30%
3	Final exam	19th	50%
	TOTAL		**100%**

*Assessment Activities (i.e., Written test, oral test, oral presentation, group project, essay, etc.).

E. Learning Resources and Facilities

1. References and Learning Resources

Essential References	• Greenberg R, Daniels S, Flanders W, Eley J, Boring J. Medical Epidemiology (LANGE Basic Science). • Textbook of Pharmacoepidemiology by Brian L. Strom, Stephen E. Kimmel, Sean Hennessy. ISBN: 9781118344866, Copyright © 2013 by John Wiley & Sons Ltd.
Supportive References	List Essential References Materials (Journals, Reports, etc.) 1- PLoS One 2- Journal of Clinical Research Bioethics 3- Journal of Urban Health 4- Journal American Medical Information Association
Electronic Materials	• Epidemiology: Study the occurrence of disease. (2002) by Thomas Koepsell and Noel Weiss. ISBN 0-19-515078-3, Oxford University Press, New York, New York. • Essential Clinical Research by Stephen P. Glasser. ISBN: 978-1-4020-8485-0 (Online), Copyright © 2008 by Springer Publishing House, Netherland. • Designing Clinical Research by Stephen B Hulley., Steven R Cummings., Warren S Browner., Deborah G Grady., Thomas B Newman (2013). ISBN/ISSN: 9781608318049, Copyright © 2013 by Wolters Kluwer. • Medical Statistics: A Textbook for the Health Sciences by Michael J. Campbell, David Machin, Stephen J. Walters. ISBN-13: 978-0470025192, (4th Edition), Copyright © 2013 by John Wiley & Sons Ltd.
Other Learning Materials	• www.science direct.com • http://www.ncbi.nlm.nih.gov/pubmed Websites of some regulatory guidelines relevant to the topics of the course.

2. Required Facilities and equipment

Items	Resources
facilities (Classrooms, laboratories, exhibition rooms, simulation rooms, etc.)	Well-equipped lecture room with at least 30-40 seats. Laboratories equipped with apparatus, instruments and laboratory facilities for laboratory tests and experiments.
Technology equipment (projector, smart board, software)	Computer room containing at least 30 systems with internet facilities.
Other equipment (depending on the nature of the speciality)	Well-equipped labs with all necessary instruments.

F. Assessment of Course Quality

Assessment Areas/Issues	Assessor	Assessment Methods
Effectiveness of teaching	Peer reviewer and HOD	Direct evaluation.
Effectiveness of Students assessment	Students	Indirect using students' questionnaires
Quality of learning resources	Faculty	Direct evaluation.

Assessment Areas/Issues	Assessor	Assessment Methods
The extent to which CLOs have been achieved	Faculty and students	Indirect using questionnaires
Other	Program Coordinator	Direct measurement of learning outcomes.

Assessors (Students, Faculty, Program Leaders, Peer Reviewers, Others (specify)

Assessment Methods (Direct, Indirect)

G. Specification Approval

COUNCIL /COMMITTEE	
REFERENCE NO.	
DATE	